

Title

Air Quality Guardian: Smart Environmental Monitoring System

Abstract

Air pollution is a major problem that affects human health and the environment. The Air Quality Guardian project monitors air quality, temperature, and humidity in real time using an ESP32 connected to sensors and an LCD display. When pollution levels exceed safe limits, the system alerts the user through the display. This project shows how embedded systems and IoT can be used to develop a low-cost air quality monitoring system for homes, classrooms, and offices.

Problem Statement

Air pollution and poor indoor air quality can cause health problems such as breathing difficulties, allergies, and fatigue. However, many people are unaware of the pollution levels around them because they lack monitoring systems.

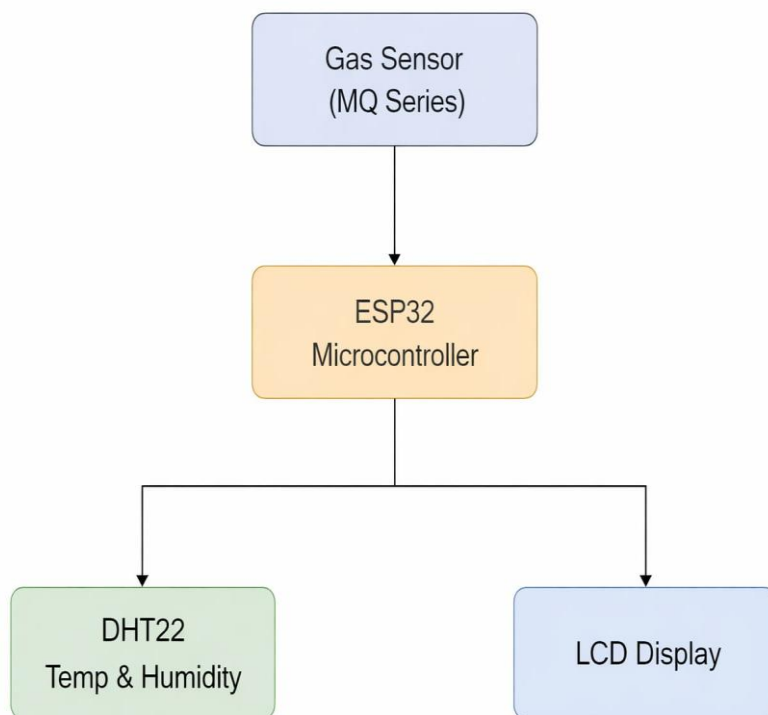
Therefore, there is a need for a simple, affordable system that can continuously monitor air quality and environmental conditions and alert users when levels become unsafe.

Objectives

The main objectives of this project are:

- To monitor air quality using a gas sensor.
- To measure temperature and humidity using a DHT sensor.
- To display environmental data in real time using an LCD screen.
- To develop a low-cost embedded monitoring system.
- To demonstrate the use of IoT and control systems in environmental monitoring

Architecture Diagram



Technologies Used

Hardware

- ESP32
- MQ Gas Sensor (Air Quality Detection)
- DHT Sensor (Temperature & Humidity)
- 16x2 LCD Display
- Breadboard
- Jumper Wires
- USB Power Supply

Software

- Arduino IDE
- Embedded C programming
- Sensor libraries

Implementation (Git)

The project implementation involves the following steps:

- Connecting sensors to the ESP32 microcontroller.
- Writing the embedded code in Arduino IDE.
- Programming the ESP32 to read sensor data.
- Processing the data and displaying it on the LCD screen.
- Testing the system for different environmental conditions.

The project code and documentation can be maintained in a repository on GitHub for version control and collaboration

Results / Output

The system successfully:

- Reads air quality levels from the gas sensor
- Measures temperature and humidity
- Displays real-time data on the LCD screen
- Provides a basic environmental monitoring system

Challenges Faced

During the development of the project, several challenges were encountered:

- Sensor calibration issues
- Difficulty uploading code to ESP32
- Library compatibility problems
- Wiring errors in the breadboard setup
- Sensor value fluctuations

These issues were resolved through troubleshooting and testing.

Future Improvements

Future enhancements for the project include:

- Mobile app monitoring using IoT
- Cloud data storage and analysis
- Air purification system integration
- Alarm or fan activation when pollution increases
- Real-time pollution alerts via smartphone
- We can put the recorded values in

